NGUYEN DUY TRANG

Open to Opportunities

Mid Mobile Engineer (Flutter, iOS, Android)

📍 Ho Chi Minh City, Vietnam • 🎂 2000 • 📞 (+84) 397 308 962 • ✉️ trangndps10349@gmail.com •
github.com/TrangDuyNguyen • 💼 linkedin.com/in/nguyen-duy-trang

PROFESSIONAL SUMMARY

OVERVIEW

Mobile Engineer with 5+ years of hands-on experience across Flutter, iOS, and Android. I have directly built and scaled social network platforms with over 1 million downloads, robust POS/Kiosk solutions across multiple industries (F&B, Retail, Spa), as well as Fintech applications (international money transfer, stock trading) and real-world aquaculture management systems. My core strengths lie in designing clean architectures, handling seamless offline data sync, and integrating peripheral hardware like thermal printers and barcode scanners. I am deeply focused on performance, ensuring butter-smooth 60 FPS rendering and zero memory leaks. Recently, I have been pioneering the integration of Multimodal Vision AI (Google Gemini) and Generative UI architectures in Flutter to craft smarter, highly interactive user experiences.

VERIFIED PRODUCTION HIGHLIGHTS

METRICS

- **1M+ Downloads:** Operating large-scale social travel network & real-time OTT on iOS & Android.
- **99.5%+ Crash-Free:** Production reliability, strict RAM lifecycle management, and locked 60 FPS.
- **Realtime & Hardware:** Low-latency WebSocket chat/calls, ESC/POS thermal printers & barcode scanners.
- **-30% Tx Time:** Optimized ordering, billing & checkout on multi-vertical POS & self-order Kiosks.
- **Offline-First & RBAC:** SQLite order sync; permission boundaries for Cashier, Warehouse, Manager.
- **226 Tests 100%:** Comprehensive automated test suite, Fastlane & GitHub Actions CI/CD pipelines.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Core:	Flutter (Dart), Swift, SwiftUI, UIKit, Kotlin, TypeScript.
Architecture & State:	Clean Architecture, Riverpod 2.x, BLoC, Provider, MVVM, SOLID, Modularization, Offline-First, RBAC.
Real-Time, Media & Data:	WebSocket, Socket.IO, RESTful APIs, SQLite, Local Caching & Sync, iOS CallKit.
Hardware & POS Modules:	Platform Channels, Thermal POS (ESC/POS), Barcode Scanner, Warehouse Management, Multi-tier Pricing.
AI Tools, CI/CD & QA:	Cursor IDE, Claude Code, Fastlane, GitHub Actions, Bitrise, Git Worktrees, Scrum/Agile.

FEATURED FLAGSHIP PROJECT

PERSONAL ARCHITECTURE

ASTROBITE — AI Food Scanner & GenUI Calorie/Macro Tracker / AI Coach

2026 – Present

Mobile Engineer & System Architect — Intelligent Nutrition & Dietary Vision Assistant

Tech Stack: Flutter 3.x, Dart 3.x, Riverpod 2.6 (Code-Gen, AsyncNotifier), AutoRoute 9.x, Google Gemini 2.0/3.8 Flash (firebase_ai), iOS WidgetKit & Android AppWidget (home_widget), Apple Health & Health Connect, Fastlane, GitHub Actions CI/CD (226 Unit/Widget Tests).

- Architected AstroBite (v2.5.1 Solar Fresh 3D Clay) based on **Feature-First Clean Architecture** with Riverpod 2.x code-gen across 10 decoupled business modules.
- Engineered camera scanner utilizing **Gemini Multimodal Vision AI (firebase_ai)** recognizing complex meals in **2–4 seconds**, estimating calories and macros (Carbs, Fat, Protein) with **0 memory leaks**.
- Pioneered **Flutter Generative UI (GenUI) Chat Cockpit** via A2UI protocol and Gemini 3.8 Flash, dynamically rendering interactive widgets (MealQuickLogCard, MacroBudgetGauge) with 1-tap Firestore logging.
- Developed **Cross-Platform OS Home Widgets** (iOS WidgetKit & Android AppWidget via home_widget) supporting instant camera scanner deep-link **< 300ms** and reactive data sync with zero background battery drain.
- Established a suite of **226 automated unit & widget tests (100% passing)** with zero flaky tests; automated multi-platform release builds and deployment using Fastlane & GitHub Actions.

NGUYEN DUY TRANG — COMMERCIAL WORK EXPERIENCE & CREDENTIALS

5+ YEARS HANDS-ON EXPERIENCE

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY

08/2025 – 08/2026

Mobile Developer — Multi-Industry POS Kiosk Ecosystem (F&B, Retail & Spa Services)

Tech Stack: Flutter, Android Kotlin, BLoC, SQLite, Platform Channels, ESC/POS Thermal Printer, Barcode Scanner, RBAC, Fastlane.

- Architected **POS & self-order Kiosk** checkouts tailored for 3 business verticals: **F&B** (table layout, kitchen ticket printing), **Retail** (barcode scanning, rapid cart), and **Spa Services** (technician booking, treatment packages), cutting **30%** transaction time.
- Designed granular **Role-Based Access Control (RBAC)** across distinct operational roles (Cashier, Warehouse Keeper, Spa Technician, Store Manager, Admin).
- Developed **Goods & Multi-tier Pricing** module: custom pricing per branch, spa treatment packages, F&B food combos, and automatic promotional rules.
- Engineered **Warehouse Management** subsystem: stock-in/stock-out receipts, inter-branch transfers, and real-time inventory tracking.
- Built comprehensive **Reporting & Analytics**: shift revenue breakdown, multi-model sales statistics (F&B, Retail, Spa), and stock movement reconciliation.
- Architected **Offline-first** order sync (BLoC, SQLite) and developed custom **Platform Channels (Kotlin)** interfacing with ESC/POS thermal receipt printers and hardware barcode scanners.
- Configured automated build and release pipelines using **Fastlane & GitHub Actions**.

HAHALOLO SOCIAL TRAVEL NETWORK

05/2020 – 06/2025 (5 years)

Mobile Developer (iOS & Flutter) — Ecosystem of 6 Apps: Social, OTT, Travel, Fintech & Outsource

Tech Stack: iOS Swift, Autolayout, VIPER, MVVM, RxSwift, Starscream, Alamofire, IG List, SQLite, CallKit, Flutter, Add-to-App, Clean Architecture, Riverpod, Firebase.

- **[Hahalolo — Travel Social Network]** Developed flagship travel social app (Tour, Flight, Hotel, Handnotes) achieving **1M+ downloads**. Built native iOS modules (Swift, Autolayout) with **VIPER** and **Cocoapods**. Integrated Maps, Deep Linking, Autoplay Video, Firebase Crashlytics. Added Flutter via **Add-to-App**. Mobile team: 20.
- **[Halome — OTT Realtime Messaging]** Engineered real-time messaging app (SSO, Anonymous) using Native iOS Swift (**MVVM**). Integrated **Starscream (WebSocket)**, **Alamofire**, **IG List**, **RxSwift**, and **SQLite** offline. Implemented **CallKit** (Audio/Video), DeepLink, multimedia/location sharing. iOS team: 5.
- **[Haloki — Fintech]** Maintained & bug-fixed international money transfer app (**192 countries**) in Flutter. Ensured stable financial transaction flows, **KYC** verification and data security.
- **[Hahalolo Invest — Fintech]** Maintained & bug-fixed stock trading app in Flutter. Real-time chart rendering, portfolio management and buy/sell order workflows.
- **[Aquannets — Outsource for Galactic Holdings]** Developed aquaculture farm management app in Flutter. **Clean Architecture + Riverpod**, SQLite offline-first at farming ponds. Mobile team: 2.
- **[SocTrip — QC & Testing]** Served as **QC**: end-to-end testing, test cases, regression testing and quality sign-off before each Store release.

EDUCATION & CREDENTIALS

ACADEMIC

FPT POLYTECHNIC COLLEGE
Associate Degree in Information Technology (Software Development Major)

English: Professional Working • **Vietnamese:** Native

ENGINEERING DISCIPLINE & METHODOLOGY

PRACTICES

Architecture & Quality: Feature-First Clean Architecture, Strict SOLID principles, Ponytail Mindset (Zero Bloat, shortest working diff).

Testing & Delivery: 100% Deterministic Testing (Domain, Data, Widget), Automated CI/CD (Fastlane, GitHub Actions, Bitrise).